

Capstone Project – Front End Development

Project Title

E-Commerce Web Application (MVP)

1. Project Overview

The objective of this capstone project is to design, develop, test, and deploy a Minimum Viable Product (MVP) for an online e-commerce application using modern web technologies.

Learners will apply the concepts covered during the course, including frontend development with React, backend development using Node.js and Express, database integration with PostgreSQL, source control using Git and GitHub, and cloud deployment using Vercel.

The completed application is intended for educational purposes and demonstrates the software development lifecycle (SDLC) from planning through deployment.

2. Project Objectives

The project aims to:

- Develop a complete full-stack web application.
 - Apply React component-based development.
 - Build RESTful APIs using Express.
 - Store and retrieve data from PostgreSQL.
 - Integrate frontend and backend applications.
 - Practice Git-based collaborative development.
 - Deploy the application to a cloud platform.
 - Produce SDLC documentation throughout the project.
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3. In Scope

The following features are included in the MVP.

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User Features

- User Registration
 - User Login
 - User Logout
 - Protected Dashboard
 - Session management using JWT
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Product Management

- View all products
 - View product details
 - Search products
 - Filter products by category (optional)
 - Display product images
 - Display product price and description
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Shopping Cart

- Add product to cart
 - Update product quantity
 - Remove product from cart
 - View cart summary
 - Calculate total price
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Checkout

- Simple checkout screen
- Order confirmation page
- Save order to database

(Payment gateway integration is not included.)

Database

The application will store:

- Users
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- Products
 - Categories (optional)
 - Shopping Cart
 - Orders
 - Order Items
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Administration

A simple administrator interface may include:

- Add product
- Edit product
- Delete product

If time is limited, these operations may be tested through API endpoints instead of a user interface.

Technical Scope

Frontend

- React
- React Router
- Fetch API
- CSS or Bootstrap

Backend

- Node.js
- Express.js

Database

- PostgreSQL

Version Control

- Git
- GitHub

Deployment

- Backend deployed on Vercel Serverless Functions
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- PostgreSQL hosted on Neon
 - Frontend deployed on Vercel
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4. Out of Scope

The following features are intentionally excluded from this MVP.

Authentication

- Forgot Password
 - Email Verification
 - Social Login
 - Multi-factor Authentication
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Shopping Features

- Wishlist
 - Product Reviews
 - Ratings
 - Coupons
 - Discounts
 - Product Recommendations
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Payments

- Credit Card Processing
 - PayPal
 - Stripe
 - Razorpay
 - UPI Integration
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Order Processing

- Shipment Tracking
 - Delivery Notifications
 - Invoice Generation
 - Return Management
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- Refund Processing
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Inventory

- Stock Alerts
 - Supplier Management
 - Warehouse Management
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User Features

- Profile Picture Upload
 - Address Book
 - Multiple Shipping Addresses
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Administration

- Analytics Dashboard
 - Sales Reports
 - User Management
 - Inventory Reports
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Performance

- Redis Cache
 - CDN
 - Image Optimization
 - Load Balancing
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Security

- OAuth Login
 - Role-Based Access Control
 - Rate Limiting
 - Audit Logs
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5. Assumptions

- Learners have completed the React, Node.js, Express, PostgreSQL, Git, and deployment modules.
 - Development will be completed within five working days.
 - Teams consist of 3–5 members.
 - Internet access is available for deployment.
 - Product images may be stored using URLs instead of file uploads.
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6. Constraints

- Duration is limited to five days.
 - Only MVP features will be implemented.
 - Third-party payment integration is excluded.
 - Only essential validations are required.
 - Focus is on functionality rather than advanced UI design.
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7. Deliverables

Each team must submit:

- Source code (GitHub Repository)
 - Frontend Application
 - Backend Application
 - PostgreSQL Database
 - API Documentation
 - ER Diagram
 - Wireframes
 - Test Cases
 - Bug Log
 - Deployment Guide
 - README
 - Final Project Report
 - Project Presentation
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8. Success Criteria

The project will be considered successful if:

- Users can register and log in.
 - Products are retrieved from the database.
 - Users can add products to the shopping cart.
 - Orders are stored successfully.
 - The frontend communicates correctly with the backend APIs.
 - Data persists in PostgreSQL.
 - The application is successfully deployed.
 - All required project documentation is submitted.
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9. Project Timeline

Day	Activities
Day 1	Planning, requirements, wireframes, database design, project setup
Day 2	Backend API development and database integration
Day 3	React frontend development and API integration
Day 4	Testing, debugging, deployment, documentation
Day 5	Final validation, presentation, demonstration, and submission
